

Pay-Equity Audit with Supervised Learning

Our role: a pay-equity regulator

Lélia Hervy · Thomas Pasquier--Guillard · Robin Cacarié

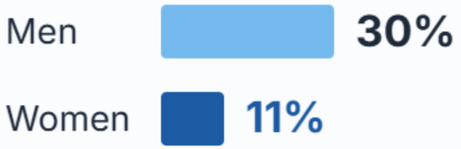
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Our role: a pay-equity regulator

Business question: which employers should we investigate, and on what evidence?

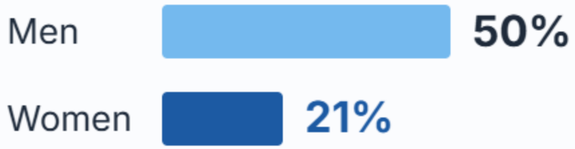
All workers

Share earning more than $50K



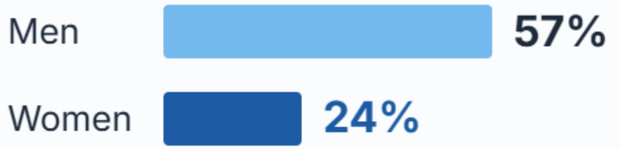
Same degree: Bachelors

Share earning more than $50K



Same job: managers

Share earning more than $50K



So what? The gap does not disappear when we compare people with the same degree or the same job. Raw gaps are not enough: under “equal pay for equal work”, we must compare each person with what their job-related profile predicts.

Adult census data, full dataset (48,842 people). Men: light blue · Women: navy.

Our data and what the model may use

USED · job-related factors

- **Age**
- **Education (years)**
- **Occupation**
- **Hours per week**
- **Workclass**
- **Investment income**

EXCLUDED · protected attributes and proxies

- ~~**Sex**~~
- ~~**Race**~~
- ~~**Native country**~~
- ~~**Relationship**~~
- ~~**Marital status**~~

> **Husband = 99.99% men: relationship encodes sex**

+5 features we created, e.g. part-time work: 28% of women vs 12% of men, a legitimate reason to earn less.

Adult census (1994) · 48,842 people · only 24% earn > $50K → we judge models on average precision, not accuracy

What the model may use

USED · JOB-RELATED FACTORS

- Age
- Education (years)
- Occupation
- Hours per week
- Workclass
- Investment income

EXCLUDED · PROTECTED OR PROXIES

- ~~Sex~~
- ~~Race~~
- ~~Native country~~
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> "Husband" = 99.99% men: relationship encodes sex.

ADDED · 5 NEW FEATURES

Works overtime (> 40 h): 40% earn > $50K vs 17%

Part-time (< 35 h): 7% vs 27%; 28% of women vs 12% of men

Has investments, net and log capital gain: 58% vs 19%

So what? The exclusions happen inside the pipeline, so sex and race can never reach the model, even in production. Part-time work is a legitimate reason to earn less, and the model now accounts for it.

Also dropped: fnlwgt (census sampling weight) and education (exact duplicate of education years).

How the model works

Employee profile
age, education, occupation, hours, workclass, investments

→

Reference pay model
trained on the population, never sees sex or race

→

Expected probability
of earning more than $50K for this profile

For each group: **actual rate − expected rate = unexplained gap**

1 · No protected attributes
otherwise the model learns the discrimination and hides it

2 · Good ranking
the expected rate must follow real differences in qualifications

3 · Calibrated probabilities
the expected rate is an average of probabilities, so they must be right

A negative gap means the group earns less than its qualifications predict: a reason to investigate, not a verdict.

Three models, one winner

Average precision on the validation set, after tuning (main metric: focuses on the rare > $50K class; random = 0.24)

Logistic Regression

0.692

Simple and transparent, but one linear score misses interactions.

Random Forest

0.752

Biggest jump: the value of a degree depends on the job and on age.

LightGBM · selected

0.776

Best ranking and recall, trains in about 1 second.

Imbalance: class weights

Same gain as undersampling or SMOTE, but keeps every real person and creates no synthetic ones.

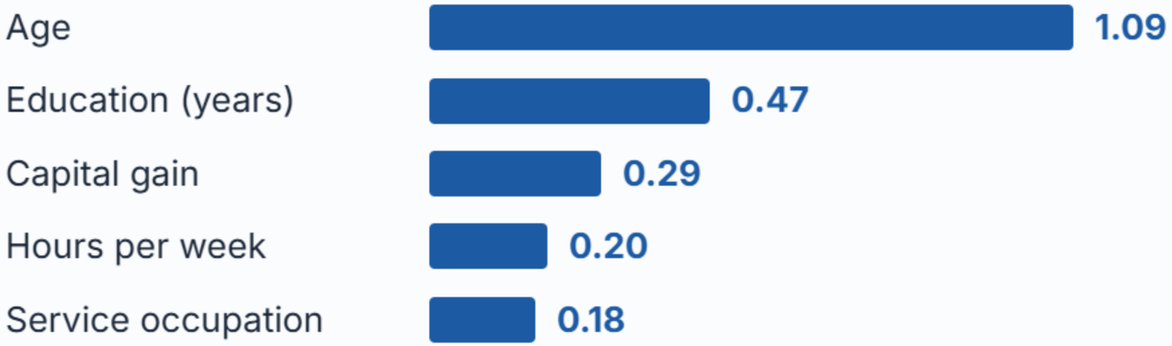
Tuning: Optuna + 5-fold CV

Only +0.005 AP: the choice of model family mattered far more than its settings.

One scikit-learn Pipeline: drop excluded columns, add 5 features, impute, scale, one-hot, model · every run logged in MLflow

What drives the predictions

Mean |SHAP value|: average impact on a prediction



ONE PREDICTION EXPLAINED

Age 50, professional degree, self-employed. Predicted **91%** chance of earning more than $50K, but earns less.

It looks like underpayment, yet there is no employer to blame.

→ **Flags are leads for inspectors, not verdicts.**

So what? The model prices a profile the way a pay grid would: experience, qualification, investment income, working time and job. These are factors an employer cannot dispute.

SHAP (TreeExplainer) on 2,000 validation people. SHAP describes what the model uses, not what causes income.

Can we trust the probabilities, and act on them?

1 · FIX THE PROBABILITIES

Average predicted chance of earning > $50K



Class weights inflated every probability. Left as is, **every group would look about 13 points underpaid.** Calibration error: 0.135 → 0.009.

2 · DECIDE WHEN TO FLAG

Flag an employee when the calibrated chance is ≥ 50% (“more likely than not”) but they earn less

82%

When the model expects > $50K, it is right 82% of the time. The F1-optimal threshold (0.39) would drop this to 68%: more false alarms for each inspector.

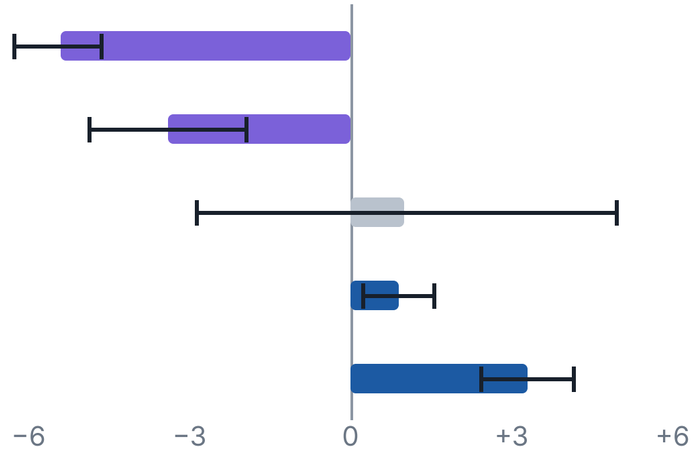
So what? Calibration makes the comparison between groups fair. A strict threshold keeps inspectors on strong cases, because their time is the scarce resource.

Isotonic calibration learned on out-of-fold training predictions · threshold chosen on validation · test set untouched until the end

Two groups earn less than their profile predicts

ACTUAL VS EXPECTED SHARE EARNING > $50K · GAP IN POINTS, 95% CI · GREY = NOT SIGNIFICANT

GROUP	ACTUAL	EXPECTED	GAP [95% CI]
Women	11.1%	16.5%	**−5.4** [−6.3, −4.6]
Black	13.3%	16.7%	**−3.4** [−4.9, −1.9]
Asian-Pac.	26.4%	25.4%	+1.0 [−2.9, +5.0]
White	25.3%	24.4%	+0.9 [+0.2, +1.6]
Men	30.4%	27.1%	**+3.3** [+2.4, +4.2]



WHAT IT MEANS

Women: about **176 of 3,259** "missing" above $50K. 1 in 3 whose profile predicts > $50K does not get it (men: 1 in 7).

Black employees: gap spread over moderate profiles, visible for the group, rarely in individual flags.

Men +3.3: the mirror image, since the reference is the population average.

So what? Two gaps whose 95% range stays clearly below zero: grounds to open investigations. Unexplained is not proven: the employer gets the chance to justify the gap.

Untouched test set (9,769 people) · 1,000 bootstrap resamples · groups under 100 people not interpreted

Hiding sex from the model is what reveals the gap

SAME TUNED LIGHTGBM, CALIBRATION AND THRESHOLD	BEFORE: ALL COLUMNS incl. sex, race, relationship, marital status	AFTER: JOB-RELATED ONLY (OURS) pre-processing mitigation
ROC-AUC (performance)	**0.930**	**0.893**
Demographic parity difference, sex	0.187	**0.103**
Equalized odds difference, sex	0.114	**0.042**
Unexplained gap, women	**+0.2**	**−5.4**
Unexplained gap, Black employees	**+1.1**	**−3.4**

So what? The full model is more accurate *because* it has learned the discrimination, so every employer would look fair. Our fairness definition is **equal pay for equal work**, measured by the unexplained gap; the classic metrics improve too.

Rejected: post-processing (ThresholdOptimizer). Group-specific thresholds would erase the very signal we measure, and need sex and race at prediction time.

Untouched test set · only the input columns differ between the two models

Four checks: the result holds, and is conservative

1 · NEW PEOPLE

0.893

ROC-AUC on validation *and* test

Calibration error 0.009 on people never seen before.

→ **our choices did not overfit**

2 · BLIND TO SEX

65.2%

same profile, man or woman, any race

Identical expected probability: sex and race never reach the model.

→ **the reference is neutral**

3 · LEAKY COLUMNS

−6.6

gap for women without capital income

Capital gains are part of total income. Without them: Black −3.8.

→ **conclusion holds, gaps even grow**

4 · HIDDEN PROXIES

0.79

ROC-AUC to guess sex from our inputs

Occupation and hours still carry sex: unequal access to good jobs is “explained away”.

→ **our gaps are a lower bound**

So what? If anything, the audit under-states the problem. That is the right side to err on when the result can lead to an investigation.

All checks on the untouched test set · gaps in percentage points

From model to tool: screening, not sentencing

FASTAPI SERVICE + WEB INTERFACE · POST /AUDIT

Employees audited	Actually earn > $50K	Expected to earn > $50K	Potentially underpaid
2,000	**24.1%**	**24.2%**	**44**
real people from the test set	483 people	average probability from the model	2.2% of employees

Unexplained gap = actual share of high earners – share expected from the job profile, in percentage points. A **negative** gap means the group earns less than its profile predicts. It is **significant** when its 95% confidence interval does not cross 0. It is a reason to investigate, not a proof of discrimination.

By sex

Groups sorted from the largest shortfall to the largest surplus.

Group	Employees	Actual	Expected	Unexplained gap (points, 95% CI)	Gap	Underpaid
Female	641	10.1%	14.9%		**−4.8** [−6.6, −2.9]	8
Male	1,359	30.8%	28.6%		**+2.1** [+0.4, +3.8]	36



Significant shortfall · Significant surplus · Not significant · Faded rows: fewer than 100 employees, not reliable

An inspector sends a company's records and gets the gap per group, its 95% range and the flagged employees. Here, one simulated company of 2,000 test-set employees: women −4.8 [−6.6, −2.9].

RECOMMENDATION

Screen every company's records. Open an investigation when a group's gap is significantly negative; use the flags to choose which files to review.

LIMITS

Unexplained is not proven: seniority, region and performance are not in the data.
1994 census of individuals, not companies; total income above $50K, not salary.
Calibration checked overall, not yet group by group.

Code, report and slides: github.com/rcacarie-png/Adult-Income-Prediction



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Thank you for listening

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